

Voicing in Czech children's sibilants: children's production and adults' perception

Tanja Kocjančič^{a*} and Tomáš Bořil^a

^aInstitute of Phonetics, Charles University, Prague, Czech Republic

Institute of Phonetics, Faculty of Arts, Charles University, nám. Jana Palacha 1/2, 116
38 Praha 1, Czech Republic

Tanja.KocjancicAntolik@ff.cuni.cz

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This study examines how Czech-speaking children mark voicing in sibilants and how adult listeners perceive these contrasts in children's speech. Productions of /s, z, ʃ, ʒ/ by 31 children aged 3;5–6;5 show that voicing is cued by both duration and presence of f₀: voiceless sibilants are longer and contain less voicing than voiced sibilants. Development is non-linear, with substantial variability, particularly for voiced targets. Consistency improves most clearly by age 6. Phonetic context affects both duration and presence of f₀, indicating that productions depend on the elicitation material. Perception experiment involving 48 adults showed that speech-language therapists do not outperform non-professionals in sibilant detection. Listeners rely on both f₀ and duration and identify voiceless targets more reliably than voiced targets. Findings support controlling phonetic context and supplementing perceptual judgments with acoustic measures.

Keywords: sibilants; voicing; children; Czech; production; perception;

Introduction

About 50% of children entering speech and language therapy (SLT) in the Czech Republic are diagnosed with a speech sound disorder (NZIP, 2023), with frequent difficulties in the sibilants /s/, /z/, /ʃ/, and /ʒ/ (Neubauer, 2018). These sounds are fricatives that differ in place of articulation, alveolar (/s/, /z/) versus post-alveolar (/ʃ/, /ʒ/), and in voicing within each pair (Šimáčková et al., 2012). Non-standard productions are typically described as errors in place of articulation and/or voicing (Neubauer, 2018), reflecting the high motor demands: a narrow midline constriction against the alveolar ridge or palate, sustained lateral tongue contact, and controlled vocal-fold vibration. Importantly, the vibration of vocal folds causes lower intraoral pressure which directly affects the quality of frication in a sibilant (Stevens, 1997). Production of

voiced fricatives is thus particularly challenging due to the synchronous maintenance of frication and voicing.

No studies have examined voicing in Czech children's speech, and data on Czech adult sibilants are limited. An EPG data of eight Czech adults showed greater tongue–palate contact in /z/ and /ʒ/ than in /s/ and /ʃ/, along with a narrower midline groove in voiced targets (Skarnitzl et al., 2013). However, examination of voiced and voiceless sibilants in other languages has shown systematic articulatory differences that are needed for simultaneous production of frication and voicing. The adjustments can be seen in tongue posture, constriction, and tongue–palate contact. MRI studies of English fricatives report a more advanced tongue root and a larger pharyngeal area for voiced sibilants than for voiceless ones (Narayanan et al., 1995; Proctor et al., 2010).

Ultrasound data from Hungarian likewise indicate tongue-shape differences: compared with /s/, /z/ involves a lower posterior tongue and a higher anterior and mid tongue (Gráczí et al., 2020). Similar results have been obtained in EPG studies, showing a more anterior constriction for English word-initial /z/ and German /ʒ/ than for their voiceless counterparts, and greater anterior tongue–palate contact for German /z/ than /s/ (Fuchs et al., 2007; McLeod et al., 2006). As suggested by Gráczí et al. (2020), the differences in the anterior constriction possibly reflect apical (for /s/) vs. laminal (for /z/) strategies. Voiced sibilants are also reported to have a narrower constriction, a direct result of the lower intraoral pressure due to voicing (Liker & Gibbon, 2013, 2018; McLeod et al., 2006). Although all studies reported large inter-speaker variability, indicating speaker-specific adjustments, the overall evidence points to robust articulatory differences between voiced and voiceless sibilants that may also be expected in other languages and in children's speech.

Unlike articulatory characteristics, acoustic correlates of voicing are language-specific, since they relate to phonological contrast. In Czech, voicing is primarily cued by the presence or absence of f_0 (Šimáčková et al., 2012). Nevertheless, studies of modal and whispered adult Czech speech show that voiced fricatives, including sibilants, are shorter than their voiceless counterparts (Machač & Šturm, 2010; Svatošová & Bořil, 2023), a pattern also reported in other languages (e.g., English: Jongman et al., 2000; Greek: Nirgianaki, 2014) and in French 2–6-year-old children (Kehoe & Philippart de Foy, 2023).

Sibilant acquisition is relatively well studied, but research focuses mainly on voiceless sounds. The available evidence nonetheless suggests that voicing affects acquisition. A cross-linguistic review of acquisition norms from 27 languages (excluding Czech) reports /s/, /z/, and /ʃ/ as typically acquired by 4;0–4;11, and /ʒ/ by 5;0–5;11 (McLeod & Crowe, 2018). Czech lacks normative acquisition data, though these sibilants are generally treated as late-acquired; the Association of Speech-Language Pathologists of the Czech Republic states that all are mastered by 6.5 years (AKL, n.d.).

Evidence for voiced–voiceless differences in development comes from French: in 89 children aged 2;6–6;10, /s/ reached 90% accuracy across word-initial, -medial, and -final positions by age 4 years, and /ʃ/ by age 6 (Kehoe & Philippart de Foy, 2023). In contrast, /z/ reached 90% by age 5 in medial position and age 6 in initial position, and /ʒ/ reached 90% by age 6 in initial and medial positions. In the word-final position, voiced sibilants were not yet acquired by 6 years of age.

Speech sound disorder (SSD) is the most common pediatric speech impairment, with prevalence estimates of 2.3%–24.6% among English-speaking children aged 4–6 years (Wren et al., 2016). Reported rates are higher in Slovak, a language closely

related to Czech: SSD was found in 60% of 4-year-olds and 40% of 6-year-olds (Buntova & Guthová, 2016). The observed errors may reflect articulatory limitations and/or phonological difficulties. Czech sibilants are vulnerable under both accounts: they are articulatorily complex and require acquisition of contrasts in place (e.g., /s/ vs. /ʃ/) and voicing (e.g., /s/ vs. /z/). Errors resulting from incomplete acquisition of the voicing contrast are typically described as the phonological processes of fricative voicing and devoicing. For Czech-speaking children, there are no data on the prevalence of these processes or the age at which they are typically suppressed. Evidence from other languages is also limited, as inappropriate use of voicing has been studied mainly in stops (e.g., final devoicing and prevocalic voicing in Australian English; James, 2001). However, fricative devoicing has been documented in Brazilian and European Portuguese: among 50 4 – 8-year-old children with SSD, 54% devoiced /z/ and 52% devoiced /ʒ/ (Mota et al., 2012), and among 26 4 – 7-year-old children with SSD, fricative devoicing was one of the most prominent phonological processes (Martins et al., 2021). Difficulties with sibilant production have been documented in other languages, including English (McLeod et al., 2013; Shriberg et al., 2010; Smit, 1993), Polish (Minczakiewicz, 2017), Macedonian (Jancheska, 2024), and Croatian (Farago et al., 1998).

Assessment of sibilants should include place and manner of articulation and, in languages where relevant, voicing. In clinical SLT, assessment typically relies on SLTs' auditory judgments, often using a binary scale (correct vs. error) and error classification. Perceptual studies show that clearly correct productions of fricatives and stops are easiest to rate, whereas small deviations are most difficult (Jing & Grigos, 2022; Klein et al., 2012). Listener experience may also shape judgments. Compared to inexperienced listeners, SLTs show higher intrarater reliability and a clear preference

for the correct productions (Munson et al., 2012). Years of SLT practice do not have a clear influence on perceptual judgments. When rating place contrasts on 9-point scales (/s/–/θ/, /s/–/ʃ/, /t/–/k/, /d/–/g/), more experienced SLTs used endpoints most often, while less experienced SLTs were most accurate for /s/–/θ/ and /s/–/ʃ/ pairs (Meyer & Munson, 2021). However, in a study of 30 SLTs (2–40 years of experience) rating children’s word productions containing various error types (including voicing errors), years of practice did not predict reliability, but agreement was again highest for correct targets and lower for words with only a single error than for those with two errors (Jing & Grigos, 2022).

The study by Jing and Grigos (2022) provides more detailed evidence on voicing perception: across all error types, the 30 SLTs were most consistent when rating voicing errors. This aligns with classic findings that voicing is a highly salient cue for listeners (Miller & Nicely, 1955). Although evidence specific to sibilants is limited, available perceptual studies suggest that voicing judgments for alveolar and postalveolar sibilants depend primarily on the presence of voicing and, secondarily, on segment duration: more voicing and shorter duration are associated with voiced sibilants (Cho & Giavazzi, 2008; Stevens et al., 1992; Pape et al., 2015). For English and European Portuguese, even partial voicing can be sufficient for voiced identification: Cho and Giavazzi (2008) report high perceptual salience of even small amounts of voicing, and Pape et al. (2015) show that the presence of voicing in 25% of sibilant duration already triggers a voiced response. At the same time, these studies confirm that tokens with expected voicing patterns are identified with high accuracy (Cho & Giavazzi, 2008; Pape et al., 2015). Pape et al. (2015) further report that when voicing is present in 25–50% of the sibilant, listeners rely more on duration: shorter sibilants are more often labelled voiceless and longer ones voiced.

Current study

The reviewed literature indicates that sibilant production difficulties are widespread across languages, that Czech adults cue sibilant voicing via the presence of f0 and duration, and that SLTs reliably perceive voicing in accurate productions but show reduced reliability when errors are present. Whether these findings generalize to Czech children's speech remains unknown. The present study therefore aimed (i) to examine Czech children's production of voicing in the sibilants /s/, /z/, /ʃ/, and /ʒ/ and (ii) to compare voicing perception in experienced adult listeners (SLTs) and inexperienced adult listeners.

Method

Experiment 1: Children's production

Participants

Thirty-one typically developing monolingual Czech children (13 girls, 16 boys), aged 3;5–6;5, participated. They were grouped by age: 3-year-olds (3;5–3;11; M = 3;9; n = 3), 4-year-olds (4;0–4;10; M = 4;9; n = 8), 5-year-olds (5;0–5;11; M = 5;7; n = 13), and 6-year-olds (6;0–6;7; M = 6;3; n = 7). Parent questionnaires indicated typical development with no known impairments or a diagnosed speech sound disorder.

Data collection

Data were collected as part of a larger study on children's production of the sibilants /s, z, ʃ, ʒ/ and the trill /r/; only the sibilant data are analyzed here. Each sibilant was elicited in syllable-onset position in six disyllabic words (24 targets total: 21 word-initial, 3 in the onset of the second syllable). Targets were followed by vowels /i, a, u/ (representing the three extreme tongue positions in the Czech vowel system) or by consonants /p/ or /b/, /t/ or /d/, and /k/, /g/ or /ʒ/, which differ in tongue involvement (no tongue

involvement, anterior tongue involvement, and posterior/mid-tongue involvement, respectively). Following consonants matched the target in voicing. The complete word set is listed in Table 1.

Table 1. Word set used for data collection.

	/i/	/a/	/u/	/p; b/	/t; d/	/k; g, j/
/s/	sirup /sirup/ <i>syrup</i>	salát /sala:t/ <i>salad</i>	sudy /sudɪ/ <i>barrels</i>	spánek /spa:nek/ <i>sleep</i>	stopy /stopɪ/ <i>footprints</i>	skála /ska:la/ <i>rock</i>
/z/	zima /zima/ <i>winter</i>	zámek /za:mek/ <i>castle</i>	zuby /zubɪ/ <i>teeth</i>	zbytky /zbitkɪ/ <i>leftovers</i>	Zdeněk /zdeɲek/ [personal name]	pozdě /pozje/ <i>late</i>
/ʃ/	šípy /ʃi:pi/ <i>arrows</i>	šaty /ʃatɪ/ <i>dress</i>	šunka /ʃuŋka/ <i>ham</i>	šperky /ʃperkɪ/ <i>jewelry</i>	štrúdl /ʃtru:dl/ <i>strudel</i>	škola /ʃkola/ <i>school</i>
/ʒ/	židle /ʒidle/ <i>chair</i>	žába /ʒa:ba/ <i>frog</i>	župan /ʒupan/ <i>bathrobe</i>	dlažba /dlaʒba/ <i>tiles</i>	každý /kaʒdi:/ <i>every</i>	ždímá /ʒji:ma:/ <i>squeeze</i>

Children produced four repetitions in an imitative picture-naming task with a prerecorded model produced by a native Czech-speaking adult female (Edwards & Beckman, 2008). Ultrasound and audio data were recorded simultaneously (Articulate Instruments Ltd., 2012) using a probe-stabilization headset (Articulate Instruments Ltd., 2008). Recordings took place in the children's kindergartens.

Data analysis

Target sibilants were first manually segmented and transcribed. To be included in the final data set, the target sibilant had to be realized as a fricative with a place of articulation between interdental and velar. This means that some of the productions were marked as having an error in the place of articulation; however, the places are not extreme for a Czech speaker and are within the scope of standard speech sounds or a common developmental error (e.g. interdental). We believe that such criteria created a

set of sounds with comparable articulatory, particularly tongue, precision that allowed evaluation of voicing. The final dataset included 2565 tokens (90% of all productions).

Two measures were extracted for each token: duration and percentage of voicing (PrctVoiced). PrctVoiced was quantified by measuring f0 (in Praat; Boersma & Weenink, 2025) across the token and calculating the proportion of each voiced part; fully voiced sibilants had a value of 100, and fully voiceless ones of 0.

Measures were analyzed within age groups and by target voicing and following segment type (consonant vs. vowel), yielding four contexts: voiceless+C, voiceless+V, voiced+C, and voiced+V. Statistical analyses were conducted in R (R Core Team, 2025) using lme4 (Bates et al., 2015) and emmeans (Lenth & Piaskowski, 2025) packages. Duration was modelled using a linear mixed-effect model with age group, target voicing, and following segment type as fixed effects with interaction among them. As random effects, we had intercepts for a child (speaker) and a word pronounced. To model the relationship between the intended production of the target voicing (voiced or voiceless) with duration and the PrctVoiced, we fitted a logistic regression using a generalized linear mixed-effect model (glmer, binomial family). Fixed effects were duration and PrctVoiced with the interaction term, and we included the intercept of the child (speaker) as a random effect.

Experiment 2: Adults' perception

Participants

Forty-eight native Czech adults participated in the perception experiment: 31 non-professional listeners (19 women, 11 men; 18–70 years; median 27.5, IQR 22–42) with diverse professional backgrounds, and 17 speech-language therapists working with

children (15 women, 2 men; 26–62 years; median 28, IQR 27–35). Both groups included speakers from Bohemia, Moravia, and Silesia.

Data collection

All participants completed an online four-alternative forced-choice identification task in PsyToolkit (Stoet, 2010, 2017). Stimuli were isolated sibilant segments extracted from the children’s productions. The test comprised 104 items ranging from 0 to 100 PrctVoiced in /s–z/ and /ʃ–ʒ/ continua. Each sibilant contributed five endpoint tokens (0% for /s, ʃ/; 100% for /z, ʒ/) originally produced before a voicing-matched consonant (20 items total). The remaining 84 items (42 per continuum) had intermediate voicing ($0\% < \text{PrctVoiced} < 100\%$); 46 came from vowel contexts and 38 from consonant contexts, though only the sibilant portion was presented.

After hearing each item, listeners had to decide whether they heard /s/, /z/, /ʃ/, or /ʒ/. In instructions, they were asked to evaluate their first impression (emphasizing that there is no “right” answer), but one additional replay was allowed. Participants completed the task independently using headphones. For each listener, the items were presented in a random order to prevent the hysteresis effect. The order of the four buttons with answers was also random for each participant (some listeners may prefer, e.g., the first button in case of uncertain answers). The experiment included a six-item practice block and a main phase split into three blocks with two music-filled breaks.

Data analysis

For each intended sibilant, we calculated confusion matrices of adult listeners’ responses, quantifying correct identifications and substitutions involving voicing and/or place of articulation. To model the relationship between the perceived voicing (voiced or voiceless) with the duration and the PrctVoiced, we fitted a logistic regression using a

generalized linear mixed-effect model (glmer, binomial family). Fixed effects were duration, PrctVoiced, and their interaction, with random intercepts for child (speaker) and listener.

Results

Experiment 1: Children's production

Median durations (Table 2) indicate that voicing affects sibilant duration across age groups: in all groups, children produced longer voiceless than voiced sibilants. The difference was greatest in the youngest group (alveolars: 60 ms; post-alveolars: 62 ms). From ages 3 to 6 years, voiceless targets shortened (−20 ms for the alveolar pair; −14 ms for the post-alveolar pair), whereas voiced targets lengthened (+16 ms for both pairs). The change is, however, not linear: 5-year-olds showed longer durations than 4-year-olds for all four targets. Duration was also influenced by the following segment, with both voiced and voiceless sibilants longer before vowels than before consonants in all age groups.

Table 2. Median and interquartile range of sibilant duration (ms) for each target, and for subgroups defined by target voicing (voiced, voiceless) and following segment type (C, V). Values are reported separately for each of the four age groups.

	3-year-olds	4-year-olds	5-year-olds	6-year-olds
/s/	188 (34)	168 (57)	181 (96)	168 (71)
/ʃ/	188 (47)	166 (48)	173 (91)	174 (64)
/z/	128 (37)	142 (66)	147 (67)	144 (68)
/ʒ/	126 (27)	124 (58)	142 (75)	142 (56)

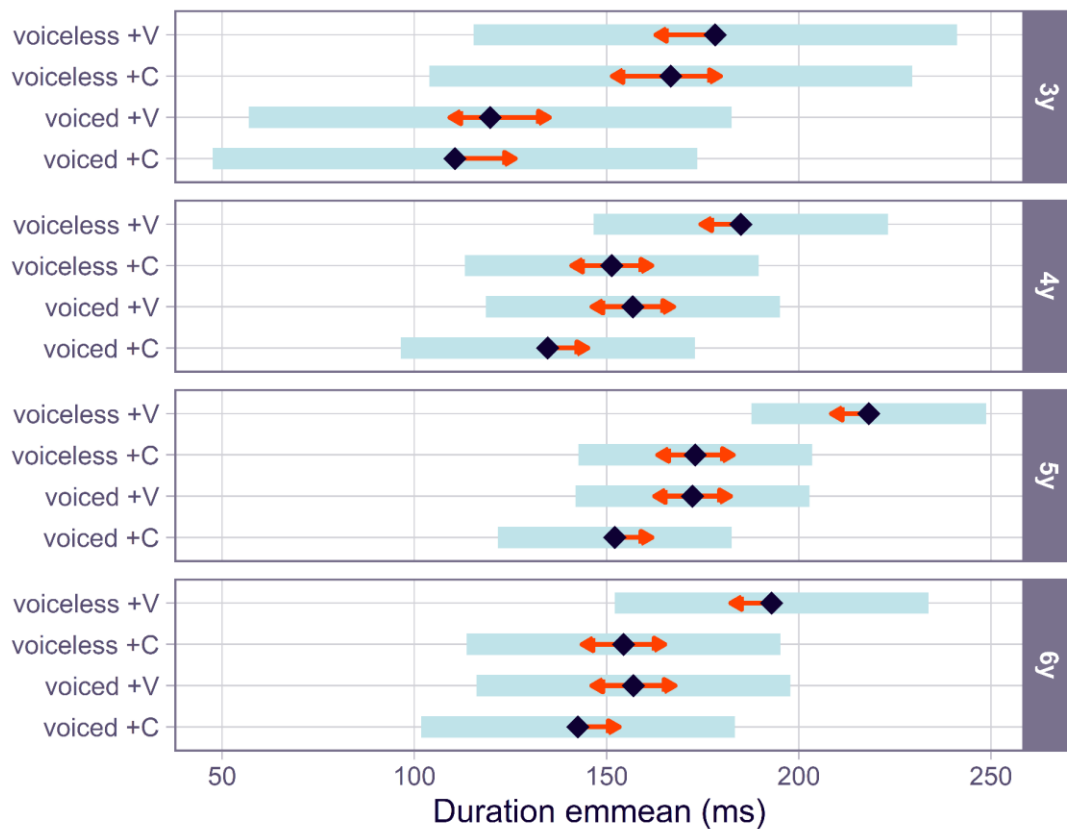
Voiceless +V	193 (35)	182 (48)	193 (84)	186 (64)
Voiceless +C	181 (42)	149 (58)	156 (76)	148 (64)
Voiced +V	128 (34)	145 (67)	144 (87)	148 (73)
Voiced +C	123 (30)	130 (64)	143 (61)	140 (58)

Figure 1 shows the analysis of the linear mixed-effect model fitted to the produced duration of sibilants using estimated marginal means. P-values were Tukey-adjusted for a family of four comparisons within each age group.

The analysis revealed that for 3-year-olds, voiceless sibilants were longer than voiced sibilants in both vowel and consonant contexts ($p < 0.0001$), with no significant effect of the following segment. For 4- and 5-year-olds, voiceless+V was longer than voiceless+C (4y: $p = 0.0005$; 5y: $p < 0.0001$) and voiced+V (4y: $p = 0.0046$; 5y: $p < 0.0001$) and voiced+V was longer than voiced+C (4y: $p = 0.0362$; 5y: $p = 0.0406$). In addition, for 5-year-olds, voiceless+C was longer than voiced+C ($p = 0.0308$); this trend also looks promising for the 4-year-olds, but the lack of significance is probably due to the smaller size of this group. Overall, in the 4- and 5-year-old groups, duration was influenced by both voicing and the following segment. For 6-year-olds, only voiceless+V was longer than the other subgroups ($p \leq 0.0002$), suggesting a limited effect of both voicing and following segment type.

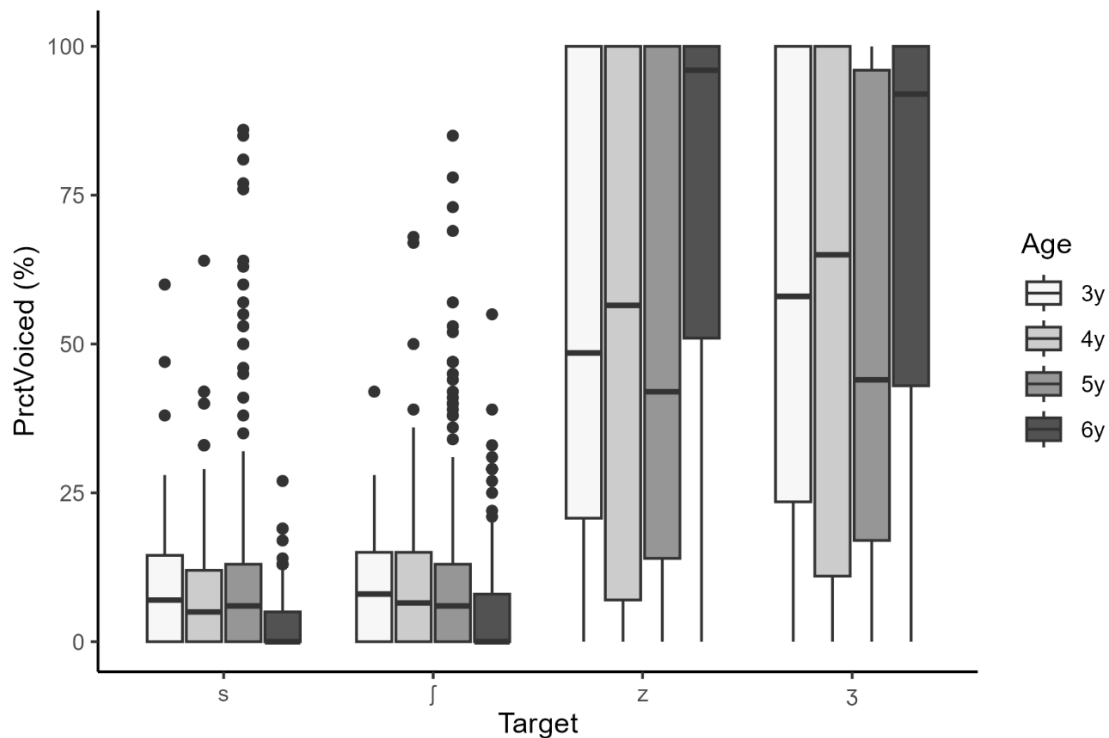
Figure 1. Estimated marginal means (x-axis) of sibilant production duration (ms) by age group, split by target voicing (voiced, voiceless) and following segment type (C, V) (y-axis). Arrows denote confidence intervals suitable for pairwise comparisons (non-

overlapping arrows indicate a significant difference). Bars show confidence intervals around each estimated mean and are not intended for between-group comparisons.



PrcVoiced (Figure 2) showed that voiceless targets were typically produced with little or no voicing. Across age groups, 46% of /s/ and 42% of /f/ were fully voiceless; no token exceeded 86% voiced duration, and only 4% had voicing in more than 50% of the segment. Voiced targets exhibited greater variability (0–100% PrcVoiced). Across age groups, 36% of /z/ and 35% of /ʒ/ were fully voiced, and 42% of voiced tokens contained voicing for less than half of their duration. Median PrcVoiced indicated age related differences. In 6-year-olds, medians were 0% for both voiceless targets and close to 100% for voiced ones (92% for /z/ and 96% for /ʒ/). In the three younger groups, medians were 5–8% for voiceless targets and 42–65% for voiced.

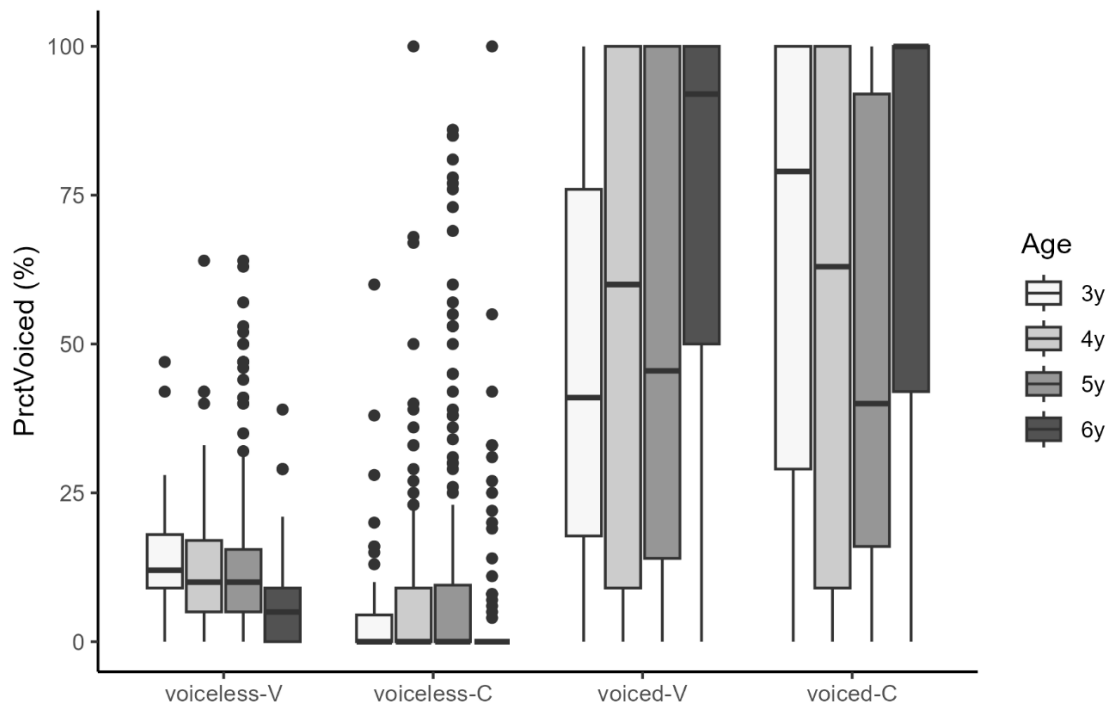
Figure 2. Percentage of voicing (PrctVoiced) for the four target sibilants in the 3-, 4-, 5-, and 6-year-old groups.



PrctVoiced was also influenced by the following segment (Figure 3). For voiceless targets, voiceless+C had lower median PrctVoiced than voiceless+V; in the +C condition, the median was 0% in all age groups. For voiced targets, the +C condition yielded higher median PrctVoiced in two groups (3- and 6-year-olds), with 6-year-olds reaching a median of 100% when targets were followed by a consonant.

Consistent with this, /s/ was fully voiceless in most +C tokens (67%, 71%, 64%, 94% across the four age groups) but in fewer +V tokens (5%, 18%, 10%, 39%). The same pattern held for /ʃ/: fully voiceless tokens were more frequent in +C (77%, 61%, 55%, 84%) than in +V (0%, 11%, 17%, 33%). As is evident from Figure 3, the amount of fully voiceless productions increases with age. Differences between +C and +V were smaller for voiced targets. /z/ was fully voiced in 48%, 37%, 25%, and 47% of +C tokens versus 19%, 41%, 28%, and 53% of +V tokens; /ʒ/ was fully voiced in 41%, 50%, 22%, and 54% of +C tokens versus 24%, 37%, 26%, and 38% of +V tokens.

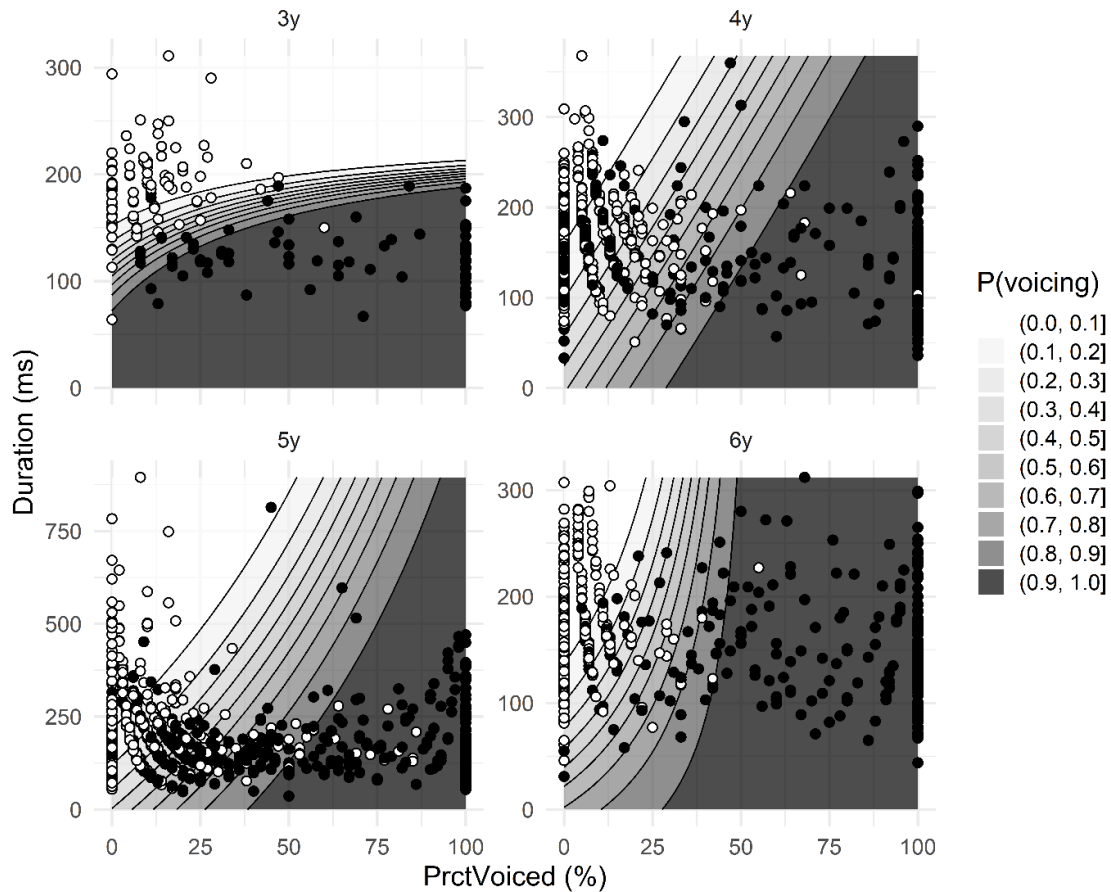
Figure 3. Percentage of voicing (PrctVoiced) in voiceless and voiced targets followed by a vowel (+V) or a consonant (+C) in the 3-, 4-, 5-, and 6-year-old groups.



The relationship between the intended production of the target voicing (voiced or voiceless), the duration, and the PrctVoiced using the logistic regression is depicted in Figure 4. With age, there is a noticeable shift in the trend from the horizontal orientation of the boundary among the youngest (voiceless much longer than voiced and the PrctVoiced in voiced is spread almost evenly across the entire range) to a more vertical orientation in the older (utilizing much more of the PrctVoiced regardless of duration).

In all four age groups, both PrctVoiced and duration fixed effects in the model are statistically significant ($p < 0.0001$) but the interaction term between them is not ($p > 0.05$) which yields to conclusion that the curved boundaries between voiceless and voiced sibilants could be simplified to a straight line (p-values were obtained by likelihood ratio tests of the full model with the effect or interaction term in question against the model without the effect/interaction term in question).

Figure 4. Logistic regression of intended target voicing as a function of percentage of voicing (PrctVoiced) and duration, shown separately by age group. Black dots represent voiced targets and white dots voiceless targets; gray bands show the predicted probability of voicing (dark = voiced, light = voiceless).



Experiment 2: Adults' perception

The percentages of successful recognition and substitutions of intended sibilants are shown in the confusion matrices in Table 3 (non-professional listeners) and 4 (speech and language therapists). Overall, non-professional listeners show slightly higher rates of recognizing speakers' intended categories, although "accuracy" should be interpreted cautiously because some child productions may not have realized the target characteristics. Voiced sibilants were clearly more difficult, with frequent confusions

with their voiceless counterparts, especially for /ʒ/. Substitutions of place of articulation were comparatively rare.

Table 3. Confusion matrices from the perception experiment with non-professional listeners. Each cell corresponds to one target sibilant (top: targets followed by a consonant (+C); bottom: targets followed by a vowel (+V)). Within each cell, the header indicates the intended target; the bold value in the upper-left corner gives the percentage of correct identifications; the second column shows voicing substitutions; and the second row shows place-of-articulation substitutions.

	/s/	/z/	/ʃ/	/ʒ/
+C	s 80.0% z 8.5% ʃ 10.8% ʒ 0.7%	z 65.5% s 22.6% ʒ 8.5% ʃ 3.4%	ʃ 85.8% ʒ 3.4% s 10.1% z 0.7%	ʒ 54.3% ʃ 36.6% z 6.7% s 2.4%
+V	s 79.6% z 5.1% ʃ 14.0% ʒ 1.3%	z 67.1% s 19.4% ʒ 11.6% ʃ 1.9%	ʃ 92.2% ʒ 4.3% s 3.0% z 0.5%	ʒ 44.2% ʃ 43.7% z 11.4% s 0.7%

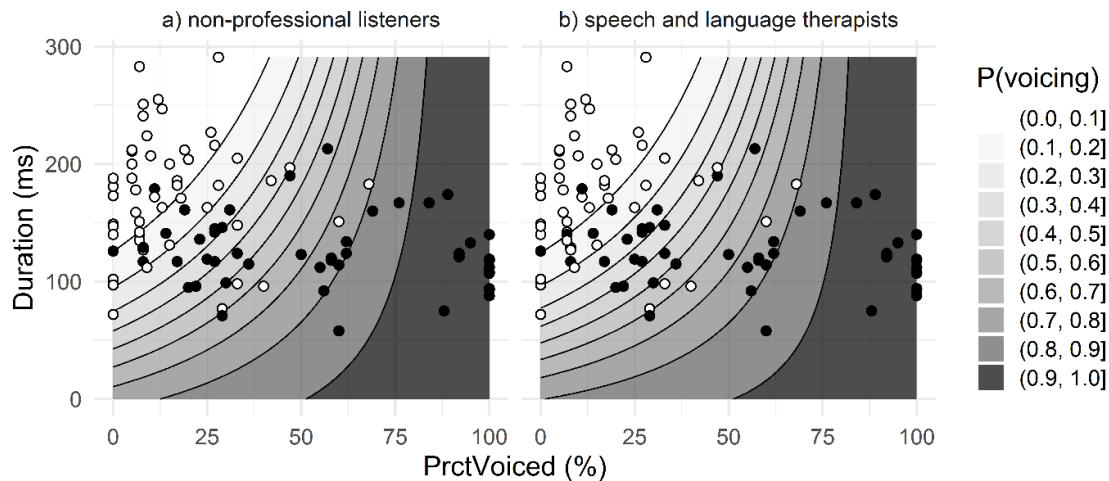
Table 4. Confusion matrices from the perception experiment with speech-language therapists as listeners. Each cell corresponds to one target sibilant (top: targets followed by a consonant (+C); bottom: targets followed by a vowel (+V)). Within each cell, the header indicates the intended target; the bold value in the upper-left corner gives the percentage of correct identifications; the second column shows voicing substitutions; and the second row shows place-of-articulation substitutions.

	/s/	/z/	/ʃ/	/ʒ/
+C	s 73.9% z 8.0% ʃ 16.0% ʒ 2.1%	z 64.0% s 23.2% ʒ 9.2% ʃ 3.6%	ʃ 85.9% ʒ 1,6% s 11.0% z 1,5%	ʒ 51.5% ʃ 36.8% z 10.3% s 1.4%

	/s/	/z/	/ʃ/	/ʒ/
+V	s 80.4% z 3.4%	z 61.8% s 21.8%	ʃ 90.2% ʒ 4.4%	ʒ 38.9% ʃ 44.8%
	ʃ 12.3% ʒ 3.9%	ʒ 14.1% ʃ 2.3%	s 4.9% z 0.5%	z 15.8% s 0.5%

Figure 5 shows logistic-regression predictions for perceived voicing (voiced vs. voiceless) as a function of duration and PctVoiced. Dots correspond to the items, and their color to their intended voicing. Predictions were highly similar for non-professional listeners (a) and SLTs (b). In both groups, the interaction terms between PctVoiced and duration fixed effects are statistically significant ($p < 0.001$), and both fixed effects are also significant ($p < 0.0001$).

Figure 5. Logistic regression of perceived voicing as a function of percentage of voicing (PctVoiced) and duration for (a) non-professional listeners and (b) speech-language therapists. Black dots represent voiced targets and white dots voiceless targets; grey bands show the predicted probability of voicing (dark = voiced, light = voiceless).



Discussion

Children's production

Czech-speaking children mark sibilant voicing through both duration and presence of f_0 : voiceless sibilants are longer and contain less presence of f_0 than voiced sibilants. The longer duration of voiceless sibilants aligns with findings for Czech adults (Machač & Šturm, 2010; Svatošová & Bořil, 2023) and French-speaking children (Kehoe & Philippart de Foy, 2023), suggesting that temporal asymmetry is a cross-speaker and cross-age characteristic rather than an adult-specific pattern. Although the duration contrast appears in all age groups, it is developmentally unstable. From ages 3 to 6 years, voiced sibilants lengthen, which reduces within-pair differences ($/s-z/$, $/ʃ-ʒ/$). This convergence implies that duration becomes a less distinctive cue with age, even though it remains systematically related to the voicing contrast.

Duration trajectories also differ by segment. For $/s/$ and $/ʃ/$, duration shortens from ages 3 to 4, increases at age 5, and then diverges at age 6: $/s/$ returns to approximately the 4-year level, whereas $/ʃ/$ remains near the 5-year level. Voiced targets show a single lengthening, but at different ages ($/z/$ around age 4; $/ʒ/$ around age 5), indicating that developmental timing is segment-specific rather than uniform across the sibilant voicing or place of articulation status. Together, these patterns support conclusions that duration is a robust cue to voicing in Czech sibilants, and that its developmental course is non-linear, with periods of refinement and periods of temporary regression or reorganization.

Presence of f_0 is another important cue in expressing the voicing status of Czech sibilants. Overall, children implement voicelessness more consistently than voicing: nearly half of voiceless targets are fully voiceless, whereas only about one-third of

voiced targets are fully voiced. The asymmetry persists beyond endpoint tokens: 96% of voiceless targets contain voicing for less than half of the segment, but only 58% of voiced targets contain voicing for more than half. Although productions are not evaluated against adult norms, these distributions indicate that sustaining voicing during frication is more difficult than suppressing it, most likely due to increased motor-control demands and aerodynamic constraints. With age, both fully voiceless targets and fully voiced targets become more frequent. The difference is most notable at age 6, particularly for voiced sibilants, and accompanied by overall reduced variability. Consequently, these changes increase the contrast between voiced and voiceless sibilants.

Both duration and PrctVoiced point to non-linear development of sibilant voicing with improvement from ages 3 to 4 and from 5 to 6, but with a reversal around age 5 (except for voiced sibilant duration). Such non-linearity is compatible with ongoing, non-uniform vocal-tract growth (Fitch & Giedd, 1999; Vorperian et al., 2005, 2009) and the resulting need for continual adjustment of speech movements and refinement of motor control (Ménard, Schwartz & Boë, 2004).

In sum, Czech children aged 3–6 use duration and f_0 to signal voicing in alveolar and post-alveolar sibilants. Duration stabilizes earlier, whereas consistent use of vocal-fold vibration remains challenging for children under 6 years of age. This further suggest that applying correct duration is an easier task for the developing motor control system than fine laryngeal–supralaryngeal coordination.

SLTs use various materials for speech sound assessment, but detailed testing of phonetic-context effects is often impractical in clinical practice. Because of this, the presented analysis examines a simplified factor: following segment type (vowel or

consonant). Sibilants are generally longer before vowels than before consonants, reflecting temporal compression observed in consonant clusters (Gilbert & Purves, 1977). However, this effect is not uniform across ages: it is significant only for 4- and 5-year-olds; 6-year-olds show it only for voiceless sibilants; and 3-year-olds show no duration adjustment to the type of the following segment.

Similarly, the amount of present voicing also does not show uniform context effects. Voiceless sibilants are on average more frequently produced as fully voiceless when followed by a voiceless consonant than by a vowel. In contrast, voiced sibilants are not reliably realized as fully voiced when followed by a voiced consonant or vowel. The results were expected for the voiceless targets where the following C was also voiceless, and V was voiced, given that latter may induce earlier onset of vocal-fold vibration, particularly in speakers with immature coordination. For voiced targets, differences between vowel and consonant contexts do not allow the same interpretation and the results suggest broader difficulty in maintaining voicing during frication.

Nevertheless, these findings indicate that in assessment of sibilant voicing, SLTs should control phonetic context as much as possible, particularly for the voiceless sibilants and ideally keep it constant across the targets. Children's productions should also be compared to age-matched norms rather than adult standards.

Adults' perception

The perception experiment compared sibilant voicing detection of non-professional listeners and SLTs and revealed that SLTs do not outperform non-professionals. It is possible that the selected task with isolated syllables was particularly challenging and it would be interesting to see whether SLTs perform better on whole words which more closely reflect clinical assessment and provide additional contextual cues. Overall,

adults find it easier to identify voiceless than voiced sibilants produced by children and make more voicing detection errors on voiced than on voiceless targets (e.g. voiced sibilants are perceived as voiceless), and more on post-alveolar than on alveolar voiced sibilants. Phonetic context does not yield systematic differences in sibilant identification, but it does call for further investigation.

Interestingly, the perception experiment revealed that tokens labelled “voiced” span the full PrctVoiced range, and tokens with the same PrctVoiced receive different responses. This indicates that voicing alone does not determine categorization and that listeners integrate multiple cues. Perception depends on both voicing and duration, consistent with findings for European Portuguese (Pape et al., 2015).

Overall, voicing in voiced sibilants is more demanding in both production and perception. Children’s use of voicing is variable and matures with age, and perceptual identification of sibilant voicing in children’s speech is not necessarily reliable. These findings support combining perceptual judgments with acoustic information when evaluating sibilant voicing in clinical or research contexts.

Clinical implications

This study has two main implications. First, it demonstrates that Czech-speaking children mark voicing via duration and f_0 . The development is non-linear, but duration control stabilizes earlier. Voiced sibilants are more difficult to produce, and this should be reflected in the assessment criteria. Additionally, sibilant voicing is influenced by the following segment and SLTs should consider the elicitation material, particularly the phonetic context of target sounds. Second, perception of voicing, especially for voiced sibilants, is challenging for adults, and the judgments also depend on both f_0 and duration. We therefore recommend supplementing perceptual assessment with acoustic

measures (e.g., f0 tracking in Praat; Boersma & Weenink, 2025) when evaluating sibilant voicing.

Limitations and future research

Several limitations should be addressed in future work. First, the age groups were uneven in size and age range; larger, more balanced samples may reveal additional developmental patterns, particularly around age 5, and would allow analyses of potential sex differences. Second, sibilant production should also be examined in word-medial and word-final positions, consistent with typical clinical assessment protocols. Third, perception of isolated sibilants should be compared with perception in words and nonwords to better approximate clinical testing materials.

Conclusions

This study investigated voicing in sibilants produced by 3- to 6-year-old Czech-speaking children, and perception of these sibilants by adult SLTs and non-professional listeners. The results show that sibilant voicing utilizes temporal and laryngeal cues, both in production and in perception. Children's voiceless /s, ʃ/ are longer and show minimal voicing, whereas voiced /z, ʒ/ are shorter and exhibit greater voicing. However, development is non-linear and highly variable. Control of duration stabilizes earlier, while sustained presence of voicing remains inconsistent until the age of 6 years. Although not systematically, phonetic context influences both cues. In perception, SLTs do not perform better than non-professional listeners in sibilant identification. Adult listeners identify voiceless targets more reliably than voiced targets and make more errors on voiced sibilants, especially /ʒ/. In clinical practice, voicing assessment should control phonetic context, interpret productions relative to age norms, and, where possible, incorporate acoustic analysis to support perceptual decisions.

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